

Database Technologies

Assignment 7 (Optional)

1. Display department name and manager first name.

```
D6_96342_Jagdish@localhost> SELECT d.department_name, e.first_name FROM departments d INNER JOIN employees e ON d.manager_id = e.employee_id;
```

department_name	first_name
Administration	Jennifer
Marketing	Michael
Purchasing	Den
Human Resources	Susan
Shipping	Adam
IT	Alexander
Public Relations	Hermann
Sales	John
Executive	Steven
Finance	Nancy
Accounting	Shelley

11 rows in set (0.001 sec)

2. Display department name, manager name, and city.

```
D6_96342_Jagdish@localhost> SELECT d.department_name, e.first_name, l.city FROM departments d INNER JOIN employees e ON d.manager_id = e.employee_id LEFT OUTER JOIN locations l ON d.location_id = l.location_id;
```

department_name	first_name	city
Administration	Jennifer	Seattle
Marketing	Michael	Toronto
Purchasing	Den	Seattle
Human Resources	Susan	London
Shipping	Adam	South San Francisco
IT	Alexander	Southlake
Public Relations	Hermann	Munich
Sales	John	Oxford
Executive	Steven	Seattle
Finance	Nancy	Seattle
Accounting	Shelley	Seattle

11 rows in set (0.001 sec)

3. Display country name, city, and department name.

```
D6_96342_Jagdish@localhost> SELECT c.country_name, l.city, d.department_name FROM departments d LEFT OUTER JOIN locations l ON d.location_id = l.location_id LEFT OUTER JOIN countries c ON l.country_id = c.country_id;
```

country_name	city	department_name
United States of America	Seattle	Administration
Canada	Toronto	Marketing
United States of America	Seattle	Purchasing
United Kingdom of Great Britain and Nort	London	Human Resources
United States of America	South San Francisco	Shipping
United States of America	Southlake	IT
Germany	Munich	Public Relations
United Kingdom of Great Britain and Nort	Oxford	Sales
United States of America	Seattle	Executive
United States of America	Seattle	Finance
United States of America	Seattle	Accounting
United States of America	Seattle	Treasury
United States of America	Seattle	Corporate Tax
United States of America	Seattle	Control And Credit
United States of America	Seattle	Shareholder Services
United States of America	Seattle	Benefits
United States of America	Seattle	Manufacturing
United States of America	Seattle	Construction
United States of America	Seattle	Contracting
United States of America	Seattle	Operations
United States of America	Seattle	IT Support
United States of America	Seattle	NOC
United States of America	Seattle	IT Helpdesk
United States of America	Seattle	Government Sales
United States of America	Seattle	Retail Sales
United States of America	Seattle	Recruiting
United States of America	Seattle	Payroll

27 rows in set (0.001 sec)

4. Display job title, department name, employee last name, starting date for all jobs from 2011 to 2015.

```
D6_96342_Jagdish@localhost> SELECT j.job_title, d.department_name, e.last_name, jh.start_date FROM employees e LEFT OUTER JOIN jobs j ON e.job_id = j.job_id LEFT OUTER JOIN departments d ON e.department_id = d.department_id RIGHT OUTER JOIN job_history jh ON e.employee_id = jh.employee_id WHERE YEAR(jh.start_date) BETWEEN 2011 AND 2015;
```

job_title	department_name	last_name	start_date
Marketing Manager	Marketing	Martinez	2014-02-17
Administration Vice President	Executive	Garcia	2011-01-13
Administration Assistant	Administration	Whalen	2012-07-01
Administration Vice President	Executive	Yang	2011-10-28

4 rows in set (0.001 sec)

5. Display job title and average salary of employees.

```
D6_96342_Jagdish@localhost> SELECT j.job_title, AVG(e.salary) FROM employees e RIGHT OUTER JOIN jobs j ON j.job_id = e.job_id GROUP BY(j.job_title);
```

job_title	AVG(e.salary)
Public Accountant	8300.000000
Accounting Manager	12008.000000
Administration Assistant	4400.000000
President	24000.000000
Administration Vice President	17000.000000
Accountant	7920.000000
Finance Manager	12008.000000
Human Resources Representative	6500.000000
Programmer	5760.000000
Marketing Manager	13000.000000
Marketing Representative	6000.000000
Public Relations Representative	10000.000000
Purchasing Clerk	2780.000000
Purchasing Manager	11000.000000
Sales Manager	12200.000000
Sales Representative	8350.000000
Shipping Clerk	3215.000000
Stock Clerk	2785.000000
Stock Manager	7280.000000

19 rows in set (0.001 sec)

6. Display job title, employee name, and the difference between maximum salary for that job and salary of the employee.

```
D6_96342_Jagdish@localhost> SELECT j.job_title, e.last_name, j.max_salary-e.salary AS salary_difference FROM employees e INNER JOIN jobs j ON e.job_id = j.job_id;
```

job_title	last_name	salary_difference
Public Accountant	Gietz	700.00
Accounting Manager	Higgins	3992.00
Administration Assistant	Whalen	1600.00
President	King	16000.00
Administration Vice President	Yang	13000.00
Administration Vice President	Garcia	13000.00
Accountant	Faviet	0.00
Accountant	Chen	800.00
Accountant	Sciarra	1300.00
Accountant	Urman	1200.00
Accountant	Popp	2100.00
Finance Manager	Gruenberg	3992.00
Human Resources Representative	Jacobs	2500.00
Programmer	James	1000.00
Programmer	Miller	4000.00
Programmer	Williams	5200.00
Programmer	Jackson	5200.00
Programmer	Nguyen	5800.00
Marketing Manager	Martinez	2000.00
Marketing Representative	Davis	3000.00
Public Relations Representative	Brown	500.00
Purchasing Clerk	Knoo	2400.00
Purchasing Clerk	Baida	2600.00
Purchasing Clerk	Tobias	2700.00
Purchasing Clerk	Himuro	2900.00
Purchasing Clerk	Colmenares	3000.00
Purchasing Manager	Li	4000.00
Sales Manager	Singh	6000.00
Sales Manager	Partners	6500.00
Sales Manager	Errazuriz	8000.00
Sales Manager	Cambrault	9000.00
Sales Manager	Zlotkey	9500.00
Sales Representative	Tucker	2000.00
Sales Representative	Bernstein	2500.00
Sales Representative	Hall	3000.00
Sales Representative	Olsen	4000.00
Sales Representative	Cambrault	4500.00
Sales Representative	Tuvault	5000.00
Sales Representative	King	2000.00

7. Display the last name and job title of employees who earn a commission but do not belong to any department.

```
D6_96342_Jagdish@localhost> SELECT e.last_name, j.job_title, e.commission_pct, d.department_id FROM employees e RIGHT JOIN jobs j ON e.job_id = j.job_id LEFT JOIN departments d ON e.department_id = d.department_id WHERE ISNULL(d.department_id);
```

last_name	job_title	commission_pct	department_id
Grant	Sales Representative	0.15	NULL

1 row in set (0.001 sec)

8. Display details of jobs that were done by any employee who is currently drawing more than 15000 of salary.

```
D6_96342_Jagdish@localhost> SELECT j.job_id, j.job_title, e.last_name, e.salary FROM employees e LEFT OUTER JOIN jobs j ON e.job_id = j.job_id WHERE e.salary > 15000;
```

job_id	job_title	last_name	salary
AD_PRES	President	King	24000.00
AD_VP	Administration Vice President	Yang	17000.00
AD_VP	Administration Vice President	Garcia	17000.00

3 rows in set (0.001 sec)

9. Display department name, manager name, and salary of the managers, who have the experience of more than 12 years.

```
D6_96342_Jagdish@localhost> SELECT d.department_name, e.employee_id, e.last_name, e.hire_date, e.salary FROM departments d LEFT JOIN employees e ON d.manager_id = e.employee_id WHERE TIMESTAMPDIFF(YEAR, e.hire_date, NOW()) >= 12;
```

department_name	employee_id	last_name	hire_date	salary
Administration	200	Whalen	2013-09-17	4400.00
Marketing	201	Martinez	2014-02-17	13000.00
Purchasing	114	Li	2012-12-07	11000.00
Human Resources	203	Jacobs	2012-06-07	6500.00
Public Relations	204	Brown	2012-06-07	10000.00
Executive	100	King	2013-06-17	24000.00
Finance	108	Gruenberg	2012-08-17	12000.00
Accounting	205	Higgins	2012-06-07	12000.00

8 rows in set (0.001 sec)

10. Display employee name if the employee joined before his manager.

```
D6_96342_Jagdish@localhost> SELECT e1.last_name as emp_name, e2.last_name as mgr_name, e1.hire_date as emp_hire_date, e2.hire_date as mgr_hire_date
FROM employees e1 INNER JOIN employees e2 ON e1.manager_id
= e2.employee_id WHERE e1.hire_date < e2.hire_date;
```

emp_name	mgr_name	emp_hire_date	mgr_hire_date
Garcia	King	2011-01-13	2013-06-17
Williams	James	2015-06-25	2016-01-03
Gruenberg	Yang	2012-08-17	2015-09-21
Faviet	Gruenberg	2012-08-16	2012-08-17
Li	King	2012-12-07	2013-06-17
Kaufling	King	2013-05-01	2013-06-17
Marlow	Frapp	2015-02-16	2015-04-10
Ladwig	Vollman	2013-07-14	2015-10-10
Rajs	Mourgos	2013-10-17	2017-11-16
Davies	Mourgos	2015-01-29	2017-11-16
Matos	Mourgos	2016-03-15	2017-11-16
Vargas	Mourgos	2016-07-09	2017-11-16
King	Partners	2014-01-30	2015-01-05
Sully	Partners	2014-03-04	2015-01-05
McEwen	Partners	2014-08-01	2015-01-05
Ozer	Cambrault	2015-03-11	2017-10-15
Bloom	Cambrault	2016-03-23	2017-10-15
Fox	Cambrault	2016-01-24	2017-10-15
Smith	Cambrault	2017-02-23	2017-10-15
Bates	Cambrault	2017-03-24	2017-10-15
Abel	Zlotkey	2014-05-11	2018-01-29
Hutton	Zlotkey	2015-03-19	2018-01-29
Taylor	Zlotkey	2016-03-24	2018-01-29
Livingston	Zlotkey	2016-04-23	2018-01-29
Grant	Zlotkey	2017-05-24	2018-01-29
Johnson	Zlotkey	2018-01-04	2018-01-29
Sarchand	Frapp	2014-01-27	2015-04-10
Bull	Frapp	2015-02-20	2015-04-10
Bell	Vollman	2014-02-04	2015-10-10
Everett	Vollman	2015-03-03	2015-10-10
Walsh	Mourgos	2016-04-24	2017-11-16
Feeney	Mourgos	2016-05-23	2017-11-16
OConnell	Mourgos	2017-06-21	2017-11-16
Whalen	Yang	2013-09-17	2015-09-21
Jacobs	Yang	2012-06-07	2015-09-21
Brown	Yang	2012-06-07	2015-09-21
Higgins	Yang	2012-06-07	2015-09-21

37 rows in set (0.001 sec)

11. Display the employee's name, job title for the jobs, employee did in the past where the job was done less than 12 months.

```
D6_96342_Jagdish@localhost> SELECT e.employee_id, e.last_name, j.job_title, TIMESTAMPDIFF(MONTH, jh.start_date, jh.end_date) as no_of_months FROM e
employees e LEFT OUTER JOIN jobs j ON e.job_id = j.job_id RIGHT OUTER JOIN job_history jh ON e.employee_id = jh.employee_id WHERE TIMESTAMPDIFF(MONTH
, jh.start_date, jh.end_date) < 12;
```

employee_id	last_name	job_title	no_of_months
122	Kaufling	Stock Manager	11
176	Taylor	Sales Representative	9
176	Taylor	Sales Representative	11

3 rows in set (0.001 sec)

12. Display employee name and country in which he is working.

```
D6_96342_Jagdish@localhost> SELECT e.last_name, c.country_name FROM employees e LEFT JOIN departments d ON e.department_id = d.department_id LEFT JOIN locations l ON d.location_id = l.location_id LEFT JOIN countries c ON l.country_id = c.country_id;
```

last_name	country_name
King	United States of America
Yang	United States of America
Garcia	United States of America
James	United States of America
Miller	United States of America
Williams	United States of America
Jackson	United States of America
Nguyen	United States of America
Gruenberg	United States of America
Faviet	United States of America
Chen	United States of America
Sciarr	United States of America
Urman	United States of America
Popp	United States of America
Li	United States of America
Khoo	United States of America
Baida	United States of America
Tobias	United States of America
Himuro	United States of America
Colmenares	United States of America
Weiss	United States of America
Fripp	United States of America
Kaufling	United States of America
Vollman	United States of America
Mourgos	United States of America
Nayer	United States of America
Mikkilineni	United States of America
Landry	United States of America
Markle	United States of America
Bissot	United States of America
Atkinson	United States of America
Marlow	United States of America
Olson	United States of America

13. Display department name, average salary and number of employees with commission within the department.

```
D6_96342_Jagdish@localhost> SELECT d.department_name, AVG(e.salary), COUNT(e.employee_id) FROM employees e RIGHT JOIN departments d ON e.department_id = d.department_id WHERE !ISNULL(e.commission_pct) GROUP BY d.department_name;
```

department_name	AVG(e.salary)	COUNT(e.employee_id)
Sales	8955.882353	34

1 row in set, 1 warning (0.001 sec)

14. Display the count of employees, month and the year in which 3 or more employees joined in any department located in 'Seattle'.

```
D6_96342_Jagdish@localhost> SELECT MONTH(e.hire_date) AS month, YEAR(e.hire_date) AS year, COUNT(e.hire_date) AS count FROM employees e LEFT JOIN departments d ON e.department_id = d.department_id LEFT JOIN locations l ON d.location_id = l.location_id WHERE l.city = 'Seattle' GROUP BY MONTH(e.hire_date), YEAR(e.hire_date) ORDER BY COUNT(e.hire_date) DESC LIMIT 1;
```

month	year	count
9	2015	3

1 row in set (0.001 sec)

15. Display employee name, job title, start date, and end date of past jobs of all employees with commission percentage null.

```
D6_96342_Jagdish@localhost> SELECT e.last_name, j.job_title, jh.start_date, jh.end_date FROM job_history jh LEFT JOIN employees e ON jh.employee_id = e.employee_id LEFT JOIN jobs j ON jh.job_id = j.job_id WHERE ISNULL(e.commission_pct);
```

last_name	job_title	start_date	end_date
Yang	Public Accountant	2007-09-21	2011-10-27
Yang	Accounting Manager	2011-10-28	2015-03-15
Garcia	Programmer	2011-01-13	2016-07-24
Li	Stock Clerk	2016-03-24	2017-12-31
Kaufling	Stock Clerk	2017-01-01	2017-12-31
Whalen	Administration Assistant	2005-09-17	2011-06-17
Whalen	Public Accountant	2012-07-01	2016-12-31
Martinez	Marketing Representative	2014-02-17	2017-12-19

8 rows in set (0.001 sec)